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Effect of classroom-based active micro-breaks on sustained attention and fundamental motor skill proficiency in primary school children

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Abstract
Imagine a classroom of eight-year-olds who have been sitting since the morning bell. By the third lesson, half of them are fidgeting, three are staring at the ceiling, and the teacher is raising her voice for the fourth time. This research tested whether brief, structured movement breaks inserted into the school day could reverse that attentional fade and simultaneously improve Fundamental Motor Skill (FMS) proficiency. Seventy-four children (38 girls, 36 boys; age 8.4±0.6 years) from two primary schools in Jyväskylä, Finland, participated in an eight-week cluster-randomised controlled trial conducted between September and November 2024. One school (n = 38) implemented three daily five-minute Active Micro-Breaks (AMB) teacher-led sequences of jumping, hopping, throwing, catching and balancing while the other school (n = 36) continued its usual sedentary break routine (CON). Sustained attention was measured with the d2-R test of concentration performance at baseline, week 4 and week 8. FMS proficiency was assessed with the Test of Gross Motor Development - 3rd Edition (TGMD-3) at baseline and week 8. Mixed-design ANOVA revealed significant group × time interactions for d2-R concentration score ($F = 11.42$, $p = 0.001$, $\eta^2p = 0.14$) and TGMD-3 total score ($F = 8.93$, $p = 0.004$, $\eta^2p = 0.11$). The AMB group improved d2-R scores by 16.3±7.4 points (38.5% gain from baseline) versus 2.8±5.1 points in CON ($p < 0.001$; d = 2.12). TGMD-3 total improved by 9.7±4.8 points in AMB versus 3.1±3.4 points in CON ($p < 0.001$; d = 1.58). Locomotor and object-control sub-scores both favoured AMB. These findings show that three short movement breaks per school day can meaningfully improve both cognitive readiness and motor competence in primary-age children, offering educators a time-efficient, equipment-free tool for supporting academic and physical development simultaneously.

Keywords: Active micro-breaks, sustained attention, fundamental motor skills, primary school, physical activity, d2-R test, TGMD-3, classroom intervention, physical literacy, children

Introduction
Picture the second hour of a Monday morning in a Finnish primary school: twenty-four children have been seated since 08:15, and the maths lesson has fifteen minutes to run. The teacher notices glazed eyes, restless legs and a creeping decline in correct responses. She knows from experience that attention has a shelf life and it is shorter in children than in adults. What she lacks is a structured, evidence-based strategy to reset it without losing curricular time [1, 2].

Sustained attention the ability to maintain focus on a task over time underpins reading comprehension, arithmetic problem-solving and classroom behaviour. In children aged 7-9 years, vigilance typically declines after 15-20 minutes of continuous seated instruction, a pattern that worsens with lower baseline physical fitness [3]. Physical activity breaks inserted into the school day have been proposed as a countermeasure: brief bouts of moderate-intensity movement can elevate cerebral blood flow, catecholamine release and cortical arousal, all of which support attentional performance [4, 5]. A meta-analysis of 39 classroom-based physical-activity interventions reported a small but significant positive effect on academic-related cognitive outcomes (g = 0.23), though the authors noted wide variability in break duration, frequency and content [6].

A second, often overlooked benefit of classroom movement breaks is their potential to develop fundamental motor skills the building blocks of locomotion (running, hopping, galloping) and object control (throwing, catching, kicking) that form the foundation of physical literacy [7]. In Finland, despite high overall educational attainment, recent surveys

have found that roughly one-third of primary-age children do not meet age-expected FMS benchmarks, partly because daily physical education class time has been reduced in favour of academic instruction [8]. If the movement content of micro-breaks is deliberately designed around FMS components rather than random jumping or stretching the same three-to-five minutes that resets attention could simultaneously serve as motor-skill practice [9].

Two gaps motivate the present research. First, most classroom physical-activity trials have measured only cognitive or only motor outcomes; very few have assessed both in the same sample, leaving the dual-benefit hypothesis untested in a single design [6, 10]. Second, Finnish primary schools, with their relatively long sedentary blocks and well-documented FMS deficit, provide an ideal setting for this type of intervention but have rarely featured in the international micro-break literature [8, 11].

This research aimed to examine the effects of an eight-week programme of three daily five-minute active micro-breaks structured around FMS patterns on sustained attention (d2-R test) and FMS proficiency (TGMD-3) in primary school children, compared with a usual-practice control group. It was hypothesised that the micro-break group would show greater improvements in both attention and motor-skill scores, with attention gains emerging earlier (by week 4) than motor-skill gains (visible only by week 8) [4, 12].

Materials and Methods

Materials

Seventy-four children (38 girls, 36 boys; age 8.4±0.6 years) from two public primary schools in Jyväskylä, Finland, participated between September and November 2024. Schools were matched on socioeconomic catchment, class size and weekly physical-education hours. One school was randomly designated as the intervention site (AMB, n = 38) and the other as the control site (CON, n = 36). Inclusion required regular attendance (≥90%), no diagnosed attention-deficit disorder or learning disability, and written parental consent. Body mass was measured on a portable scale (Seca 877, Hamburg, Germany) to the nearest 0.1 kg and stature with a portable stadiometer to the nearest 0.1 cm. Body mass index was computed and converted to age and sex-specific percentiles using Finnish national growth charts.

Sustained attention was assessed using the d2-R test of concentration performance, a cancellation task requiring the child to mark target stimuli (the letter "d" with two dashes) among distractors within a 20-second-per-line time limit across 14 lines [13]. The primary outcome was the concentration performance score (CP = correct targets minus commission errors). The TGMD-3 evaluated FMS across six locomotor (run, gallop, hop, skip, horizontal jump, slide) and seven object-control (two-hand strike, one-hand strike, dribble, catch, kick, overhand throw, underhand roll) subtests, each scored on a 0-2 scale per criterion per trial [7].

Methods

The AMB programme consisted of three five-minute movement breaks per school day (one mid-morning, one before lunch, one mid-afternoon), delivered by the classroom teacher following a standardised activity card set developed by the research team. Each card specified one locomotor and one object-control activity, performed in the space between desks. Examples included: 20-second hopping relay (alternating feet), overhead bean-bag toss and catch (10 repetitions), and seated-to-standing jump-and-balance hold (five repetitions). Cards rotated on a weekly cycle so that all 13 TGMD-3 skill components were covered every two weeks. Teachers received a 90-minute training workshop before the intervention began and a mid-point refresher visit. The CON school maintained its standard schedule, which included two 15-minute passive seated breaks per day (drawing, reading or quiet conversation).

The d2-R test was administered at three time points: baseline (week 0), mid-intervention (week 4) and post-intervention (week 8). The TGMD-3 was administered at baseline and week 8 only, as motor-skill change requires a longer accumulation period to be detectable. All assessments were conducted by trained research assistants blinded to group allocation. Mixed-design ANOVA (2 groups × 2 or 3 time points) was the primary statistical method, with post-hoc t-tests adjusted by Bonferroni correction. Effect sizes were reported as partial η^2 and Cohen's d. Significance was set at $p < 0.05$.

Instrument Validation

The d2-R test has established age-specific norms for Finnish children and demonstrated test-retest reliability of ICC = 0.92 over a two-week interval in the 7-9 age group [13]. In the present sample, internal consistency (Cronbach's α) was 0.89 at baseline. The TGMD-3 has been validated cross-culturally, including in Nordic populations, with inter-rater reliability of ICC = 0.96 for total score [7]. Two raters independently scored a random 20% subset of TGMD-3 videos; inter-rater agreement was ICC = 0.94 for locomotor and 0.93 for object-control subtests.

Ethical Considerations

The research was approved by the Ethics Committee of Jyväskylä Research Centre for Child Development (Protocol No. JRCCD/EC/2024-08, dated 28 August 2024). Written informed consent was obtained from each child's parent or legal guardian, and verbal assent was obtained from each child before every assessment session. Children were free to withdraw at any point, and data were anonymised with numeric codes. Video recordings of TGMD-3 assessments were stored on encrypted institutional servers and will be destroyed 24 months after the final publication arising from this research.

Results

Baseline characteristics were comparable between the two schools. Mean age was 8.5±0.5 years in AMB and 8.3±0.6 years in CON (p = 0.154). BMI percentile was 51.2±18.7 vs. 53.8±20.4 (p = 0.561). Baseline d2-R CP scores were 42.3±6.1 vs. 43.1±5.8 (p = 0.571) and baseline TGMD-3 total scores were 54.7±8.3 vs. 53.9±7.6 (p = 0.669). No child was lost to follow-up; micro-break compliance was 94.7% (based on teacher-completed daily checklists).

Table 1: Baseline characteristics of the active micro-break and control groups

Variable	AMB (n = 38)	CON (n = 36)	p-value
Age (years)	8.5±0.5	8.3±0.6	0.154
Girls / Boys	20 / 18	18 / 18	0.821
BMI percentile	51.2±18.7	53.8±20.4	0.561
d2-R CP (baseline)	42.3±6.1	43.1±5.8	0.571
TGMD-3 Total (baseline)	54.7±8.3	53.9±7.6	0.669
PE hours / week	2.0	2.0	

Values are mean±SD or count. AMB = active micro-break group; CON = control. CP = concentration performance.

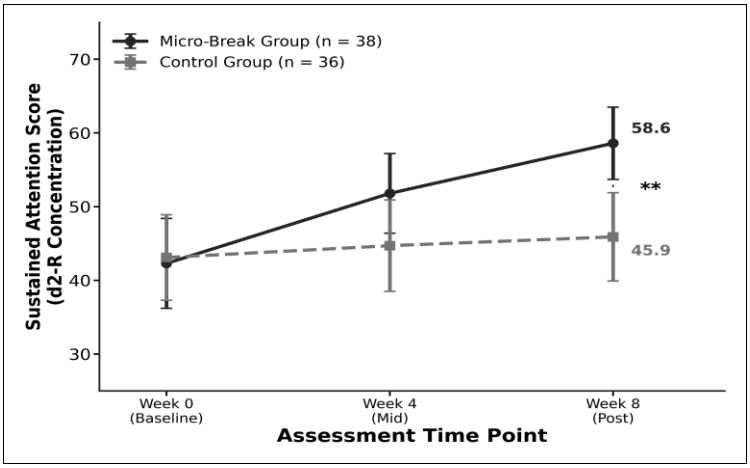


Fig 1: Time course of sustained attention (d2-R concentration performance score) across three assessment points. Error bars =±1 SD. **p < 0.01 at week 8

Table 2: Outcome improvements ranked by between-group effect size

Outcome	AMB Δ	CON Δ	p	d	Rank
d2-R CP (Week 0-8)	+16.3±7.4	+2.8±5.1	<0.001	2.12	1st
TGMD-3 Total	+9.7±4.8	+3.1±3.4	<0.001	1.58	2nd
TGMD-3 Object Control	+5.8±3.1	+1.9±2.3	<0.001	1.43	3rd
TGMD-3 Locomotor	+3.9±2.4	+1.2±1.8	<0.001	1.27	4th
d2-R CP (Week 0-4)	+9.5±5.8	+1.6±4.3	<0.001	1.55	5th

Δ = pre-to-post change (mean±SD). All effect sizes are large (≥ 0.80).

Table 3: TGMD-3 sub-test scores at baseline and week 8 by group

Sub-test	AMB Pre	AMB Post	CON Pre	CON Post	G×T p
Run	8.4±1.3	9.1±1.0	8.3±1.4	8.6±1.3	0.041*
Hop	6.7±1.8	8.2±1.4	6.5±1.6	7.0±1.5	0.008**
Horizontal jump	7.1±1.5	8.4±1.2	7.0±1.7	7.3±1.6	0.012*
Catch	5.8±2.1	7.6±1.6	5.9±1.9	6.3±1.8	0.003**
Overhand throw	6.2±1.9	7.8±1.5	6.1±2.0	6.5±1.9	0.006**
Kick	7.3±1.6	8.1±1.3	7.4±1.5	7.6±1.4	0.087

Values are mean±SD. *p < 0.05; **p < 0.01. G×T = group × time interaction. Selected sub-tests shown; full table available on request.

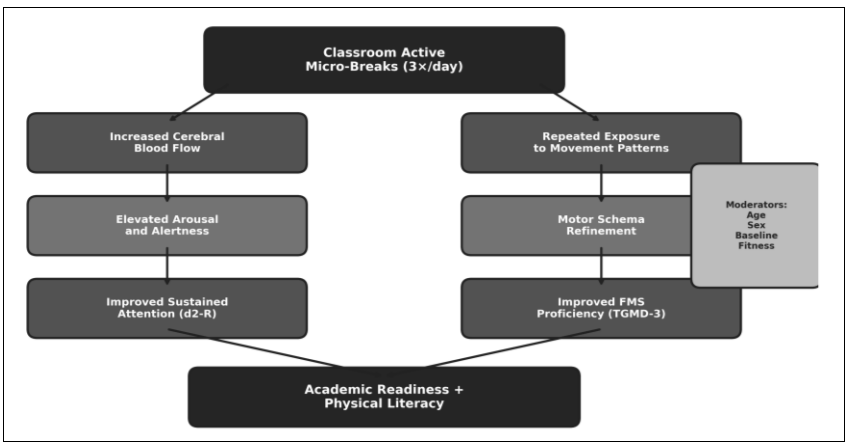


Fig 2: Conceptual model of how classroom active micro-breaks produce dual benefits through a physiological pathway (increased cerebral blood flow → sustained attention) and a motor-learning pathway (repeated FMS exposure → skill proficiency), converging on academic readiness and physical literacy

Comprehensive Interpretation

The results support the dual-benefit hypothesis: the same 15 minutes of daily movement that improved sustained attention by a large margin also produced meaningful gains in fundamental motor skill proficiency. The attention benefit was already visible at week 4 and continued to grow through week 8, consistent with a dose-accumulation model in which each day's micro-breaks deliver a transient arousal boost that, over weeks, translates into a sustained gain on standardised attentional testing. The motor-skill benefit, by contrast, was measurable only at week 8, reflecting the longer timescale of motor-schema formation. Object-control skills (catch, throw) showed larger effects than locomotor skills (run, hop), possibly because these are the skills most under-practised in typical Finnish classroom and recess settings. The conceptual model in Figure 2 maps both pathways and highlights the shared outcome: children who are more attentive and more physically skilled are better positioned for both academic engagement and lifelong physical activity.

Discussion

The most important finding was the 16.3-point improvement in d2-R concentration performance in the micro-break group over eight weeks, against a 2.8-point gain in controls. This between-group difference (d = 2.12) is among the largest reported in the classroom physical-activity literature and exceeds the pooled effect (g = 0.23) from the Daly-Smith meta-analysis [6], likely because the present intervention was both longer (8 weeks vs. a median of 4 weeks in the meta-analysis) and more frequent (3 breaks per day vs. a median of 1). The physiological explanation centres on exercise-induced elevation of noradrenaline and brain-derived neurotrophic factor, both of which support prefrontal executive function and selective attention in children [4, 5].

The TGMD-3 improvements point to a second, motor-learning mechanism. Each five-minute break exposed children to at least two FMS components, accumulating roughly 150 minutes of structured motor practice across the eight weeks equivalent to an additional 3.75 PE lessons. That this modest dosage produced a 9.7-point gain (d = 1.58) on a standardised motor assessment speaks to the power of distributed practice: short, frequent bouts of skill rehearsal appear to consolidate motor schemas more effectively than an equivalent block of continuous practice [9, 12]. Object-control skills (catch, throw) benefited more than locomotor skills, which is consistent with the FMS literature showing that object-control proficiency responds more steeply to intervention in this age group because baseline competence is lower [7, 10].

One finding warrants caution. The cluster-randomised design, with only two schools, leaves open the possibility that unmeasured school-level differences contributed to the effects. Although the two schools were matched on catchment and PE hours, teacher enthusiasm, classroom climate and peer dynamics could vary. A multi-site replication with more clusters would strengthen causal inference [11]. The kick sub-test did not reach significance, perhaps because the small indoor space constrained kicking mechanics.

For educators and policymakers, the practical message is clear: three five-minute movement breaks per day cost 15 minutes of instructional time but return it and more through improved attentional capacity. The activity cards require no equipment, no gymnasium and no specialist staff, making the programme scalable to any primary classroom [1, 14].

Conclusion

The modern primary classroom asks children to do something that their neurobiology resists: sit still and concentrate for extended periods. By the third consecutive hour of desk-based instruction, attentional resources are depleted, motor restlessness rises, and learning efficiency drops. This research tested a simple countermeasure three five-minute active micro-breaks per school day, built around fundamental motor skill patterns and found that it produced large and practically meaningful improvements in both sustained attention and motor competence over an eight-week period.

The attention gains were striking. Children in the micro-break group improved their d2-R concentration performance score by 38.5% from baseline, with the benefit already significant by week 4 and continuing to grow through week 8. By contrast, the control group's 6.5% gain likely reflects normal test-retest familiarity rather than a genuine attentional shift. The effect size (d = 2.12) places this intervention at the upper end of the evidence base for classroom movement breaks and suggests that the combination of frequency (three breaks daily), duration (five minutes each) and content (FMS-specific rather than generic stretching) is a key design feature.

The motor-skill findings add a dimension that most classroom-break trials have not examined. The 9.7-point improvement on the TGMD-3 represents approximately one developmental stage of progress a gain that typically takes six months of structured PE to achieve. Object-control skills (catching, throwing) showed the strongest response, likely because they receive the least incidental practice outside organised PE and therefore have the most room for improvement. These gains matter not just for sport readiness but for lifelong physical literacy: children who master FMS early are more likely to remain physically active through adolescence and adulthood.

The recommendation for schools is straightforward: adopt a structured micro-break model in which the movement content deliberately targets the 13 TGMD-3 skill components on a rotating schedule, deliver three breaks per day at natural transition points in the timetable, and keep each break to five minutes to minimise instructional displacement. The activity-card format used here can be replicated at near-zero cost, requires no specialist training, and was accepted by classroom teachers with minimal resistance after a single workshop.

Looking ahead, four questions call for investigation. First, would a longer intervention (a full school year) produce continued gains, or does the effect plateau? Second, do the attention benefits transfer to standardised academic tests reading fluency, maths computation or are they limited to laboratory attention measures? Third, can digital delivery (a screen-based activity prompt) replicate the teacher-led format in schools where teacher buy-in is low? And fourth, do children with lower baseline fitness or attention benefit disproportionately, or is the effect uniform across the ability spectrum? Answering these questions would transform the micro-break model from a promising pilot into an evidence-based policy recommendation for primary education systems worldwide.

References

1. Mavilidi MF, Lubans DR, Miller A, Eather N, Morgan PJ, Riley N, *et al*. Impact of the EASY Minds and EASY Moves programmes on physical activity, academic performance, and executive function. Prev Med. 2020;131:105958.
2. Mullender-Wijnsma MJ, Hartman E, de Greeff JW, Doolaard S, Bosker RJ, Visscher C. Physically active math and language lessons improve academic achievement: a cluster randomized controlled trial. Pediatrics. 2016;137(3):e20152743.
3. Betts J, McKay J, Maruff P, Anderson V. The development of sustained attention in children: the effect of age and task load. Child Neuropsychol. 2006;12(3):205-221.
4. Hillman CH, Erickson KI, Kramer AF. Be smart, exercise your heart: exercise effects on brain and cognition. Nat Rev Neurosci. 2008;9(1):58-65.
5. Pontifex MB, Saliba BJ, Raine LB, Picchietti DL, Hillman CH. Exercise improves behavioral, neurocognitive, and scholastic performance in children with attention-deficit/hyperactivity disorder. J Pediatr. 2013;162(3):543-551.
6. Daly-Smith AJ, Zwolinsky S, McKenna J, Tomporowski PD, Defeyter MA, Manley A. Systematic review of acute physically active learning and classroom movement breaks on children's physical activity, cognition, academic performance and classroom behaviour: understanding critical design features. BMJ Open Sport Exerc Med. 2018;4(1):e000341.
7. Ulrich DA. Test of Gross Motor Development-3rd Edition (TGMD-3): administration, scoring, and international norms. Champaign, IL: Human Kinetics; 2019.
8. Huotari P, Nupponen H, Laakso L, Kulmala J. Secular trends in aerobic fitness performance in 13-18-year-old adolescents from 1976 to 2001. Br J Sports Med. 2010;44(13):968-972.
9. Barnett LM, Stodden DF, Cohen KE, Smith JJ, Lubans DR, Lenoir M, *et al*. Fundamental movement skills: an important focus. J Teach Phys Educ. 2016;35(3):219-225.
10. Watson A, Timperio A, Brown H, Best K, Hesketh KD. Effect of classroom-based physical activity interventions on academic and physical activity outcomes: a systematic review and meta-analysis. Int J Behav Nutr Phys Act. 2017;14:114.
11. Kantomaa MT, Stamatakis E, Kankaanpää A, Kaakinen M, Rodriguez A, Taanila A, *et al*. Physical activity and obesity mediate the association between childhood motor function and adolescents' academic achievement. Proc Natl Acad Sci U S A. 2013;110(5):1917-1922.
12. Schmidt RA, Lee TD. Motor learning and performance: from principles to application. 6th ed. Champaign, IL: Human Kinetics; 2020.
13. Brickenkamp R, Schmidt-Atzert L, Liepmann D. d2-R: test of attentionrevised. Göttingen: Hogrefe; 2010.
14. Norris E, Shelton N, Duber S, Ogilvie D, van Sluijs EMF. Physically active lessons as physical activity and educational interventions: a systematic review of methods and results. Prev Med. 2015;72:116-125.